Chap 1

Introduction to Thermodynamics-Energy

Heat, Work and Mass

0309 3770 437/

Thermodynamics

deals with the transfer of energy from one form to another or from one place to another

Energy

is a quantitative property that must be transferred from a body or physical system to perform work or transfer heat.

Heat

is a form of energy that is transferred between two bodies due to difference in the temperature.

Energy
→ Kinetic
→ Potential
→ Chemical
→ Nuclear
→ Thermal
→ Electrical

Energy
- Macroscopic
  - ↓ K.E, P.E
- Microscopic
  - ↓ Translational, Vibrational, Rotational, Binding force } Internal Energy (U)

$$\text{Total Energy} = K.E + P.E + U$$
$$= \frac{1}{2}mv^2 + mgh + U$$

First law of thermodynamics

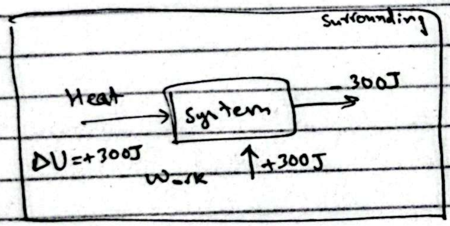
$\Delta U = Q - W$

⇒ ΔU = change in internal energy of system
⇒ Q = heat added to system
⇒ W = Work done

Mechanisms of Energy Transfer:-

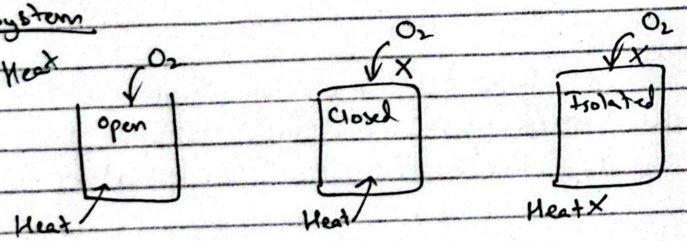
1) Work done ($W \to$ joule)
2) Heat ($q \to$ joule)
3) Mass flow

① Work



System

① Heat



$\Delta U = q - W$

⇒ $W \to$ -ive → Work done $\boxed{\text{ON}}$ the system
⇒ $W \to$ +ive → Work done $\boxed{\text{BY}}$ the system

⇒ $q \to$ +ive (absorb) → endothermic
⇒ $q \to$ -ive (release) → exothermic

Eg) 100J of work is done on the system. U increases by 74J. How much energy is tranferred as heat!

Sol

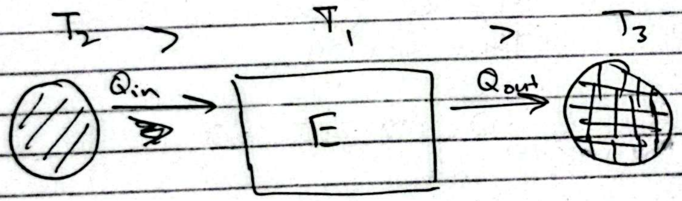
$\Delta U = Q - W$
$74 = Q - (-100)$
$74 = Q + 100$
$Q = 74 - 100$
$Q = -26J$ (exo)

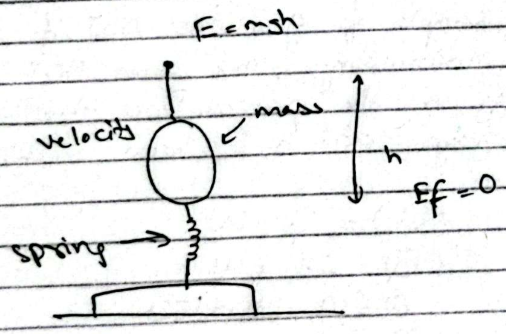
Eg) A sample of gas does 150J of work against its surrounding and loses 90J of internal energy in the process. Does the gas lose or gain heat? and how much?

Sol.

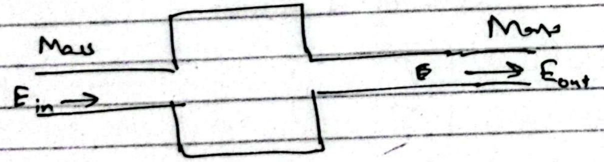
$\Delta U = Q - W$
$Q = \Delta U + W$
$Q = -90 + 150$
$Q = 60$ J (endo)

→ 1st law of thermodynamics tells us that if you add heat to a system or do work on it, the energy of system increases. Conversly, if the system does work or loses heat, its energy decreases.





③ Mass



Ch#2

Properties of Pure Substance

Pure Substance: A pure substance is defined as a material that has a homogeneous and invariable chemical composition.

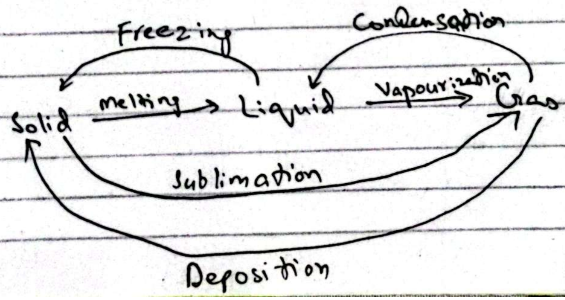
Homogenity: The substance is the same throughout, with no regions having a different composition.

Invariable Composition:- The chemical composition doesn't change even when the substance transitions between different phases.

Solid: Tightly packed molecules
Liquid: Tightly packed but moveable
Gas: Freely moveable.

Phase Changes and Diagrams

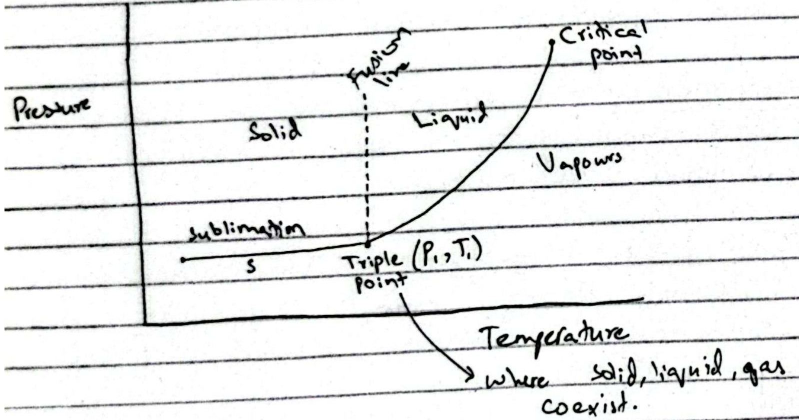
Phase changes of a material occure when it transitions from one phase/state to another.



~~Pressure $\propto$ $\frac{1}{\text{Boiling Point}}$~~

$P \propto \frac{1}{V}$ $\quad V \propto T \quad P \propto T$

Saturation Temperature: The specific temp. at which a substance undergoes a phase change under a given pressure. (vapourization, condensation)

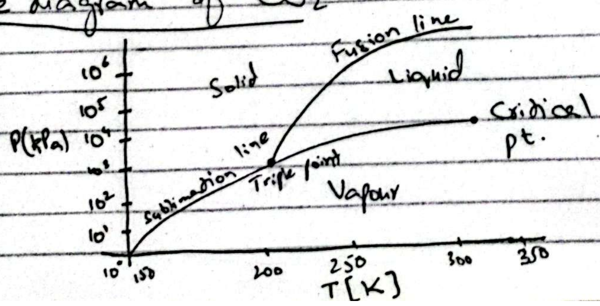


Triple point of water occurs at 0.01°C, 0.6117 kPa

Critical Point marks the end of vapourization curve on a phase diagram. Beyond this point liquid and gaseous phase becomes indistinguishable.

Critical Point of water is 374.14°C & 22.09 MPa

Phase diagram of CO_2



⇒ lowest temp. at which water can remain in liquid form: 0.01°C

⇒ If the pressure is smaller than the smallest pressure P_{sat} at a given temperature T, what is the phase?
Gas phase because the substance would be below the pressure at which it would begin to condense into liquid.

The P.V.T Surface

→ Crucial for understanding how substances behave under different conditions (P, V, T)

→ Applications: refrigerators, compressor

→ Captures the relationship P, V, T

P-V-T Surface of Water

→ ICE — has larger volume than liquid state

→ liquid water

The fusion line (solid-liquid) on pressure temp diagram has a negative slope indicating increasing the pressure lowers the freezing point.

P.V.T surface of a typical substance

The fusion line (solid-liquid) line on pressure-temp diagram has a positive slope indicating ~~the~~ increasing the pressure, increases the freezing point.

Phase Diagram

i) P-T diagram
- Fusion line
- Sublimation line
- Critical pt
- Tripple pt

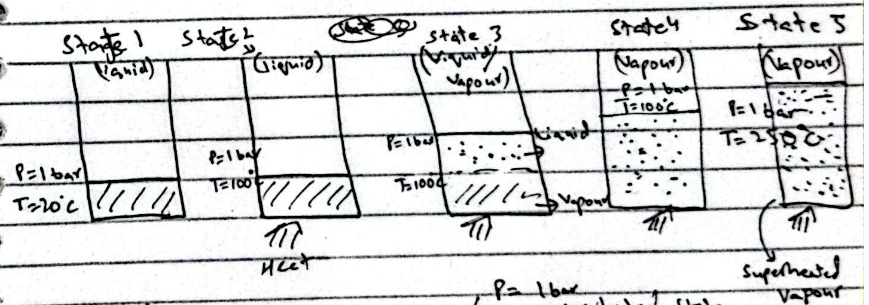
ii) T-V diagram
- Shows how volume changes with temperature at constant pressure illustrating phase change from liquid to vapour.
- Depicts the boiling phase

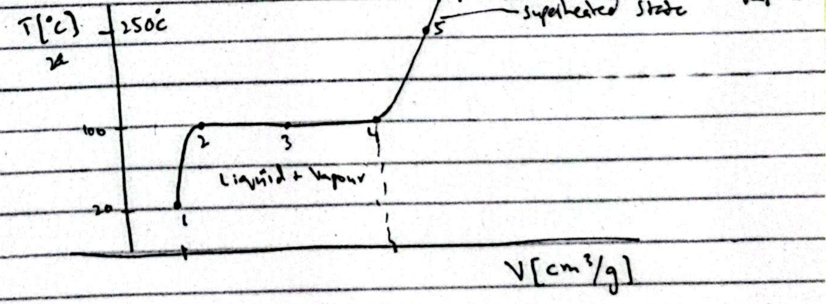
iii) P-V diagram
- Shows how pressure changes with specific volume at constant temperature.

Independent Properties

The state of substance can be defined by two independent properties. However, in the two phase diagram (during boiling or condensation), pressure and temperature are independent because they are fixed at a saturation point for a given pressure-temperature pair.

Tables of Thermodynamics Properties





Purpose ⇒ Thermodynamics Tables provide us data for various substances that describe their thermodynamic properties (like temperature, pressure, volume, specific volume, internal energy, ~~entropy~~ enthalpy and entropy).

Saturation Temperature (T_{sat}): Temperature at which phase change occurs at const P.

Saturation Pressure (P_{sat}): Pressure at which phase change occurs at const. T

Compressed Liquid — Table - 1

Saturated Liquid → Liquid about to become vapour

Saturated Liquid-Vapour mixture

Saturated Vapour → Vapour about to become liquid

Superheated Vapour — Table - 4

1 → Pressure (Table 2)
2 → Temperature (Table 3)

With two independent properties, we can find the rest:-

1) Temp (T)
2) Pressure (P)

This pair is not a group of independent properties

3) Specific Volume (V)
4) Specific Internal Energy (U)
5) Specific Enthalpy (h)
6) Specific energy (s)

F	T	P	v_f	v_{fg}	v_g
	0	P_1	v_1	v_{11}	v_{g_1}
17°C	10	P_2	v_2	v_{12}	v_{g_2}
	20	P_3	v_3	v_{13}	v_{g_3}
32°C	30	P_4	v_4	v_{14}	v_{g_4}
	40	P_5	v_5	v_{15}	v_{g_5}

- Linear interpolation
- Polynomial regression
- Moving average
- Lagrange Interpolation

Saturated Liquid and Vapour Tables

Table B1.1

v_f : Specific volume of saturated liquid
v_g : Specific volume of saturated vapour
v_{fg} : Difference b/w v_g and v_f

$$v_{fg} = v_g - v_f$$

First column : Temperature
Second column : Corresponding Saturation Pressure (P_{sat})

Table B1.2

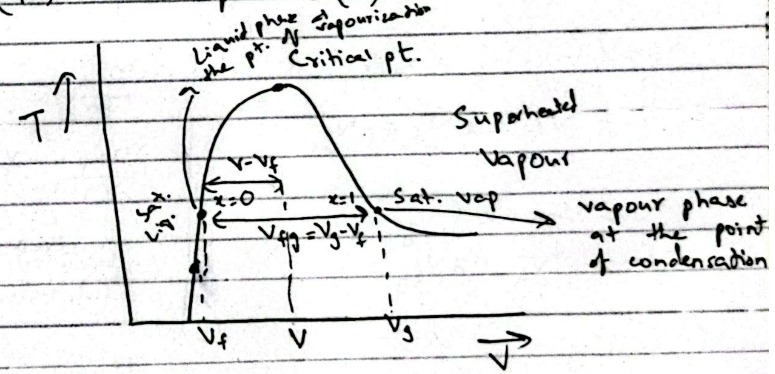
⇒ Contains the same information as Table B1.1 but organized pressure
⇒ Useful when pressure is known and temp. needs to be found.

Eg 2.1) a) $T = 120°C$ ⇒ $P = 198.5\,kPa$ ⇒ Compressed liquid
$P = 500\,kPa$ ⇒ $T = 151.86°C$ ⇒ Subcooled liquid

b) $T = 120°C$
$v_f = 0.00106 < v < v_g = 0.891$ m^3/kg
⇒ two-phase mixture of liquid & vapour

Two Phase States

A mixture of two phase of the same substance such as liquid and vapour in equilibrium at the same temperature (T) and pressure (P).



Specific Volume:-
Volume occupied by a unit mass of a substance, Measure of how much space 1kg of substance occupies.

$$v = \frac{V}{m} \implies V = m$$

Specific Volume in Two Phase:-

Total volume of two-phase mixture is the sum of the volumes of liquid (V_{Liq}) and vapour (V_{vap}).

$$V_{Total} = V_{Liq} + V_{vap} = m_{Liq} v_f + m_{vap} v_g$$

Avg. Specific Volume:-

$$v = \frac{V}{m} = \frac{m_{Liq} v_f}{m} + \frac{m_{vap} v_g}{m}$$

$$= (1-x) v_f + x v_g$$

$\left\{ \because \frac{m_{Liq}}{m} = 1 - x \right.$

$\frac{m_{vap}}{m} = x$ → ~~quantity~~ quality of substance

$$v = (1-x) v_f + x v_g$$

$$= v_f - x v_f + x v_g$$

$$v = v_f + x (v_g - v_f)$$

$$v = v_f + x v_{fg}$$

Quality:-

It is ratio of mass of vapour to total mass in a two phase mixture range $0 \le x \le 1$ $\begin{cases} x = 0 \ ; \text{ sat liq} \\ x = 1 \ ; \text{ sat vap.} \end{cases}$

Q) Calculate the specific volume of a saturated mixture of water at 200°C with a quantity of 70%. Refer Table B 1.1

Sol.

$$v = x v_{fg} + v_f = x(v_g - v_f) + v_f$$

$$= 0.7(0.12736 - 0.001156) + 0.001156$$

$$= 0.0895 \ m^3/kg$$

Compressed Liquid:- $v \approx v_f$

A liquid state at a pressure higher than the saturation pressure for a given temperature OR at a temperature lower than the saturation temp for a given pressure

~~Table 1.4~~ 1.4

$P > P_{sat}$ at a given T

$T < T_{sat}$ at a given P

$v < v_f$ at a given T or P

$u < u_f$ → Internal energy

$h < h_f$ → Enthalpy

Superheated Vapour:- $v \approx v_g$

A liquid state at a pressure lower than the saturation pressure for a given temperature OR at a temperature higher than the saturation temp for a given pressure

$P < P_{sat}$ at given T

$T > T_{sat}$ at given P

$v > v_g$ at a given T or P

$u > u_g$ → Internal energy

$h > h_g$ → Enthalpy

Ideal Gas : low pressure
high temperature
low density

~~8.2-25) T = 260°C, P = 5 MPa~~

Ideal Gas Law

Describe the behavior of gases under certain conditions. It provides simple correlation among pressure (P), temperature (T) and specific Volume (V). It is an approximation and works best under high temperature, low density and low pressure.

Ideal Gas Equation of State

$$PV = RT$$

For calculations involving mass/moles

$$PV = mRT = nRT$$

$$R = \frac{\bar{R}}{M}$$

$$\bar{R} = 8.3145 \ \frac{KJ}{Kmol.K}$$

$$PV = mRT = n\bar{R}T \qquad \therefore n = \frac{m}{M}$$

$\Rightarrow$ n is the numbers of moles.

Ideal gas law holds:-

- low pressure
- high temperature
- low density

Mass flow rate — Ideal gas law

$$P\dot{V} = \dot{m}RT = \dot{n}\bar{R}T$$

Q) Mass of air ? $M = 28.97$ kg/Kmol

Room dimensions $6m \times 10m \times 4m$

$T = 25°c$ ~~R =~~ $P = 100$ kPa

Table A.5 Pg 760

Sol.

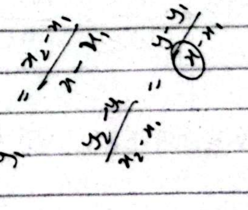
$$PV = mRT$$

$$\frac{PV}{RT} = m$$

$$\frac{(100k)(6\times10\times4)}{(0.287)(25+273)} = m$$

$m = 280.616$ kg

$$R = \frac{\bar{R}}{M} = \frac{(8.3145)}{28.97} = 0.287$$



Q) Tank has a
$V = 0.5\,m^3$
$m = 10\,kg$ of ideal gas
molecular mass = $24\,\frac{kg}{kmol}$, $T = 25°c$ $P = ?$

Sol)

$$R = \frac{\bar{R}}{M} = \frac{8.3145}{24} = 0.346$$

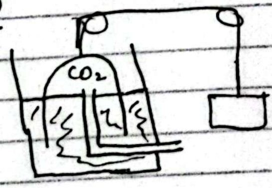
$$PV = mRT$$

$$P = \frac{mRT}{V} = \frac{(10)(0.346)(298)}{0.5}$$

$$= 2064\ KPa$$

Eg 2.8 Pg 58

⇒ A gas bell is submerged in water with its max counter balanced with ropes and pulleys. The temp is 21°C and pressure is 105 kPa. Volume increase is measured to be 0.75 m³ over a period of 185 s. What is volume flow rate and mass flow rate of flow into bell, assuming it is CO_2 gas?



Sol)
flow rate

$$\dot{V} = \frac{dV}{dt} = \frac{\Delta V}{\Delta t} = \frac{0.75}{185}$$

$$= 0.00405\,m^3/s$$

$R = 0.1889\ kJ/kg.K$

$$PV = \dot{m}RT$$

$$\dot{m} = \frac{PV}{RT} = \frac{105 \times 0.00405}{0.1889 \times (273+21)}$$

$$= 0.00766\ kg/s$$

Compressibility Factor (Z)

To account for the ~~derivates~~ deviations in ideal gas law, compressibility factor is used

$$Z = \frac{PV}{RT}$$

$$PV = ZRT$$

($Z = 1 \rightarrow$ gas is ideal)
($Z \neq 1 \rightarrow$ Non-ideal behavior, Gas is real)

→ If $Z < 1$ → Attractive forces b/w molecules dominate causing the gas to be more compressible

⇒ If $Z > 1$ → Repulsive forces dominate, gases resist compression

→ At low pressures, Z approach 1 as pressure approaches zero.

→ At high temp, for many gases like nitrogen and air, Z remains close to 1 up to pressures of around 10MPa. This means ideal gas law is still a good approximation even at moderate pressures.

Reduced Pressure

$$P_r = \frac{P}{P_c}$$; P_c : critical pressure

Reduced Temperature

$$T_r = \frac{T}{T_c}$$ T_c: critical temp.

These reduced properties allows us to plot Z as function of P_r, T_r

If $P << P_c \longrightarrow$ Gas behave nearly ideally
$P \rightarrow P_c \longrightarrow$ Gas behaves non-ideally

$\int \int \int \int$

Ch # 3

First Law of Thermodynamics for Control Mass (Energy Conversion)

→ Control mass: fixed mass system. mass does not change.
→ Control Volume: A spatial region where mass can enter & exit.
→ Heat: Energy transfer due to temperature difference.
→ Work Transfer: Energy transfer due to a force applied to a region.

Energy Equation for Control Mass

Instantaneous rate form

$$\dot{E}_{cv} = \dot{Q} - \dot{W}$$

$$\frac{\partial E_{cv}}{\partial t} = \dot{Q} - \dot{W}$$

$\dot{E}$ = Rate of total energy change within the control.

Heat (+ive) → When heat is entering the system

Work ($\dot{W}$) (+ive) → ~~on~~ work done by the system

Energy Eq. for finite Changes

Consider total energy changes ~~over~~ a process from state 1 → state 2.

- Starting point

$$\delta_{E_{cv}} = \delta Q - \delta W$$

Integration over time

$$\int_{t_1}^{t_2} \partial E_{cv} = \int_{t_1}^{t_2} \partial Q - \int_{t_1}^{t_2} \partial W$$

$$E_2 - E_1 = Q_{1-2} - W_{1-2}$$

↳ total heat added from state 1 → state 2

Energy Equation details

$$E_T = U + K\cdot E + P\cdot E$$

$$E_2 - E_1 = (U_2 - U_1) + \frac{1}{2} m (V_2^2 - V_1^2) + mg(h_2 - h_1)$$

∴ (control mass)

Q) heat 500 (absorb) W = 200 (done) Change in int. energy

$$\Delta U = q - W$$

$$\Delta U = 500 - (+200)$$

$$\Delta U = 300$$

State fnc.
Depends only on the state of the system eg U, E

Path fnc.
Depends on the specific process path taken Q, W

Pressure Volume Work

$$\delta W = p\,dv = \frac{F}{dA} \cdot dA \cdot \delta s$$

$$\delta w = F \cdot \delta s$$

$$W = nRT \ln\left(\frac{V_2}{V_1}\right) = P_1 V_1 \ln\left(\frac{V_2}{V_1}\right)$$

Iso Thermal System:-
T → same

P & V can be different.

PV = nRT
Isothermal eq. of an ~~gas~~ ideal gas

First Law of Thermodynamics — Control Mass & Cyclic Process

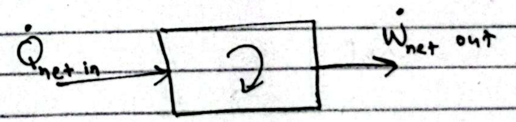
$$0 = \oint Q - \oint W$$

It implies that there is no net change in the total energy of control mass and the rate of change of energy is zero.

$$\oint Q = \oint W$$

cyclic integral of heat transfer

Cyclic integral of work done



Q) Heat absorbed = 250 kJ (from reactor)   Heat released = 150 kJ to radiator
Find net work done using 1st law of thermodynamics and efficiency?

Net work done = 250 - 150 = 100 kJ

~~η Efficiency = $\frac{Q_{out}}{Q_{in}} = \frac{150}{250} \times 100\% = 60\%$~~

$$\eta = \frac{W_{net\ out}}{Q_{net\ in}} \times 100$$

$$= \frac{100}{250} \times 100 = 40\%$$

Q) $P_1 = 300$ kPa $V_1 = 0.05\ m^3$
The gas undergoes an isothermal expansion until its volume doubles ($V_2 = 2V_1$)
During expansion, the gas absorbs 100 J of heat from surroundings?

a) $P_2 = ?$

Isothermal ⇒ T → same   hence ideal gas law

$$P_1 V_1 = P_2 V_2$$

$$\frac{P_1 V_1}{2V_1} = P_2$$

$$P_2 = 150\text{ kPa}$$

b) Calculate work done by gas during expansion?

work done by ideal gas

$$W = nRT \ln\left(\frac{V_2}{V_1}\right)$$

$$= P_1 V_1 \ln\left(\frac{V_2}{V_1}\right)$$

$$= 10.397\text{ kJ}$$

Adiabatic ⟶ No heat transfer

c) What is internal energy change (ΔU)?
For ideal gas under isothermal process
$\Delta U = 0$

Insulated material ⟶ $Q = 0$
Adiabatic process

$$\Delta U = Q - W$$
$$= -W$$

Q) Mass = 1100 kg   Kinetic energy = 400 kJ   velocity = ?

$$K.E = \frac{1}{2} m v^2$$

$$v = \sqrt{\frac{2 K.E}{m}} = 26.967\text{ m/s}$$

P.E = K.E

$$P.E = mgh$$

$$\Rightarrow \frac{K.E}{mg} = h$$

$$37.106\text{ m} = h$$

Q) $m_s = 10$ kg, $m_w = 100$ kg, $h_s = 10.2$ m, $T_s = T_w$ (State 1)
Stone ⟶ bucket, $\Delta U = ?$, $\Delta K.E = ?$, $\Delta P.E = ?$, $Q = ?$, $W = ?$
1-2 a) State 2 ⇒ stone about to enter water
2-3 b) State 3 ⇒ stone rest in water
3-4 c) State 4 → Heat transferred to surrounding such that $T_s = T_w$

a) $\Delta U = 0$, $T_s = T_w$ (isothermal)
$Q = 0$
$W = 0$

$\Delta U + \Delta K.E + \Delta P.E = Q - W$
$\Delta K.E = -\Delta P.E = -mg(h_2 - h_1)$
$= -(10)(9.8)(-10.2) = 1\,kJ$

$\Delta P.E = -1\,kJ$

b) $\Delta P.E = 0$, $Q = 0$, $W = 0$

$\Delta U + \Delta K.E = 0$
$\Delta U = -\Delta K.E = -1\,kJ$

c) $\Delta K.E = 0$, $\Delta P.E = 0$, $W = 0$, $\Delta U = -1\,kJ$

$\Delta U + \Delta K.E + \Delta P.E = Q - W$
$\Delta U = Q$
$-1\,kJ = Q$

Modes of Heat transfer

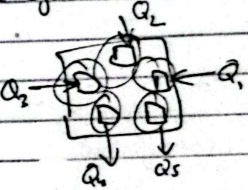
Heat: Energy transferred across the boundry of the system from one body to another (surrounding) due to temperature difference b/w two systems. Heat naturally flows from a system at high temp. to one at a low temperature. If heat doesn't flow → Thermal equilibrium

Units: Joule (SI) , BTU , Calorie

Sign Convention: Heat into the system: +ive
Heat out of the system: −ive

Path function:

$$Q_{Total} = \int \delta Q$$



① **Conduction:** Transfer of heat due to the direct contact b/w particles of matter. It happens because higher energy particle (those at higher temp.) transfer energy to lower energy molecules (those at lower temp) via molecule collisions.

~~Faraday's~~ Fourier's law of conduction

- $\dot{Q}$: Rate of heat transfer
- K : Thermal conductivity for material
- A = Cross sectional through heat is transfered
- $\frac{dT}{dx}$ = Temperature gradient

K (for metals) : about 100 W/m·K
K (for insulators) : upto 0.01 W/m.K

$$\dot{Q} = -KA \frac{dT}{dx}$$

② **Convection:** Heat transfer through a moving fluid (liquid or gas) It is a combination of bulk motion of fluid and conduction.

Newton's Law of Cooling:

$$\dot{Q} = hA\Delta T$$

convective heat tanfer coefficient

① Natural Convection

② Forced Convection

③ **Radiation:** Heat transfer due to electromagnetic waves and doesn't ~~s~~ require any medium.

e.g Heat from sun that reaches Earth.

Stefan-Boltzman Law:

$$\dot{Q} = \epsilon \sigma A T_s^4$$

- ϵ = Emissivity of the surface (0 to 1)
- σ = Stefan-Boltzman constant (5.67×10^{-8} W/m²K⁴)
- A = surface area
- T_s = Surface temp.

Problem)

Room temp = $T_{room} = 20°C$

Outside temp = $T_{ambiant} = -10°C$

Outer glass surface temp = $T_{outer} = 12.1°C$

Glass thickness : $x = 0.005$ m

$K = 1.4$ W/mK

$A = 0.5$ m²

$h = 100$ W/m²K

Conduction through Glass?

Convection from glass to ambiant air?

Sol.

$\Delta T = 20 - 12.1 = 7.9°C$

$\Delta x = 0.005$

$$\dot{Q}_{conduction} = -KA\frac{\Delta T}{\Delta x} = -(1.4)(0.5)\frac{(7.9)}{(0.005)}$$

$$= -1106 \text{ W}$$

$$\dot{Q}_{convection} = hA\Delta T$$

$\Delta T = 12.1° - (-10) = 22.1°C = 295.1$ K

$$Q_{conv} = (100)(0.5)(\cancel{295.1}\ 22.1)$$

$$= 1105 \text{ W}$$

$\frac{U - U_{fg}}{}$ $\frac{V - V_f}{V_{fg}}$

$\frac{U - U_f}{U_g}$

Extensive vs Intensive Properties

- Extensive properties → depends on the amount of matter (mass, volume, internal energy)
- Intensive property → independent of the mass of the system (temp, pressure)

Internal Energy

$$\underbrace{U}_{\text{total internal energy}} = \underbrace{m}_{\text{mass}} \underbrace{u}_{\text{specific internal energy}}$$

$U_{fg} = U_g - U_f$

Internal energy in the liquid-vapour region:-

$x = 0$ (saturated liq.)

$x = 1$ (saturated vap.)

$$U = U_{liq} + U_{vap}$$
$$= (1-x)U_f + xU_g$$
$$= U_f + x(U_g - U_f)$$

Enthalpy

Enthalpy: combines internal energy and pressure-volume work. It is used while analyzing processes that occur at constant pressure. Applications involve boilers, turbines and compressors where heat transfers at constant pressure is common (h).

Derivation:-

Consider a control mass of gas under going equilibrium constant pressure work. The work is only done at boundary of the gas state as it expands or compress. According to first law of thermodynamics,

$$U_2 - U_1 = Q_{12} - W_{12}$$

We assume a constant pressure process, the work done is given by

$$W_{12} = P(V_2 - V_1)$$

Energy Equation (First law of thermodynamics) becomes

$$U_2 - U_1 = Q_{12} - P(V_2 - V_1)$$

$$Q_{12} = (U_2 - U_1) + P(V_2 - V_1)$$

Notice that expression for heat transfer involves a combination of internal energy and pressure volume work. We define a new property called Enthalpy as:

$$\boxed{H = U + PV}$$

For constant pressure process, the heat transfer is

$$Q_{12} = H_2 - H_1$$

Specific Enthalpy: $h = u + pv$

Saturated and superheated states

h_f: enthalpy of saturated liquid

h_g: enthalpy of saturated vapour

h_{fg}: enthalpy of saturated liquid-vapour mixture

$$h = h_f + x h_{fg}$$

This is heat during the phase change of water from liquid to vapour (boiling).

Unit: kJ/kg

Applications:-

- Boilers and steam turbines

Constant pressure, the heat added

$$Q_{boiler} = H_{steam} - H_{water}$$

Work done by the turbine

$$W_{turbine} = H_{inlet} - H_{outlet}$$

- Heat Exchangers

$$Q = \dot{m} \cdot (h_{out} - h_{in})$$

$\dot{m}$ is the mass flow rate

Internal Energy $\boxed{U = h - PV}$

Enthalpy is a state property, means it only depends on the state of the substances, not on the path taken to reach that state.

Problem Analysis & Solution Technique

Steps

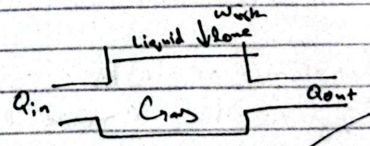
Step1 Make sketch of system

Why? Helps in understanding different aspects of system interact.

How? Indicate the direction of mass flow, heat tranfer, work and external forces

Example

Piston cylinder system: sketch the system the gas inside and allows to show any flow of heat in work done



fixed mass

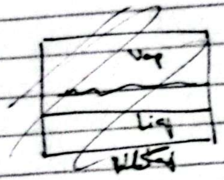
specific space through which matter flows

Step2 Define Control mass or control volume

Why? Defining boundary of system helps in applying right laws and equation.

How? Place a closed surface around the system

Q.1)



or subsystem you are analyzing. ~~That~~ Identify any in flows or out flows of mass, energy & work.

Step 3 Apply General Conservation Laws

Why? These are fundamentals to solve thermodynamics problems (conservation of energy, mass, 1st law of thermodynamics)

How? Energy eq. → ensure all inflows and outflows are mentioned

$$U_2 - U_1 = Q_{12} - W_{12}$$

Step 4 Write down any specific laws

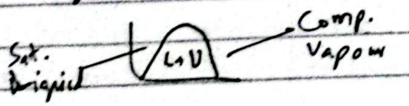
Why? These allow us to handle any specific details about the substances involved in the systems (ideal gas or steam etc)

How? Reference property tables and apply specific laws based on the nature of process (const. volume, isothermal, adiabatic).

Step 5 Solve using diagrams or tables.

Why? P-V, T-V, diagrams help visualize the process

How? P-V, T-V diagrams visualization state determination.



Step 6 Formulate and solve equations

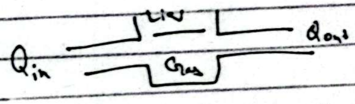
How? Combine all eq. and relationships then solve for unknown

Eq $U_2 - U_1 = Q_{12} - P(V_2 - V_1)$ Adiabatic $\hookrightarrow Q = 0$

Example) A vessel with volume $5\,m^3$ contains $0.05\,m^3$ of sat. liq water and $4.95\,m^3$ of sat. vapour at 0.1 MPa. Heat transferred to system until it is filled with sat. vapour. Determine heat transfer?

Sol

Step 1



Initial Condition: Volume liquid = $0.05\,m^3$

Volume vapour = $4.95\,m^3$

Final State : Vessel is filled with sat. vapour at some higher pressure

Step 2 Control volume and control mass

Step 3 $U_2 - U_1 = Q_{12} - W_{12}$ — As const. volume $W = 0$

$$U_2 - U_1 = Q_{12}$$

Step 4 Sat. water pressure entry (Table B.1.2)

$P = 100\,kJ$

$V_f = 0.001043\ m^3/kg$, $U_f = 417.36\ kJ/kg$

$V_g = 1.6940\ m^3/kg$, $U_g = 2506.1\ kJ/kg$

Mass of sat. Liquid

$$m_{Liq} = \frac{V_{f\,liq}}{v_f} = \frac{0.05}{0.001043} = 47.94\ kg$$

$$m_{vap} = \frac{V_{vap}}{v_g} = \frac{4.95}{1.6940} = 2.92\ kg$$

Total mass $= 50.86\ kg$

$$\eta_{thermal} = \frac{W}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

$$W = Q_H - Q_L$$

$$\beta = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L} = \frac{1}{\frac{Q_H}{Q_L} - 1}$$

Conservation of Mass

"Mass cannot be created or destroyed"

- Control volume but not control mass

For control volume, any change in mass inside the control volume must be accounted for any mass entering or leaving through the control surface. The equation of continuity is

$$\frac{dm_{cv}}{dt} = \sum \dot{m}_{in} - \sum \dot{m}_{out}$$

if $\frac{dm_{cv}}{dt} = 0$, then mass remains constant.

Mass Flow Rate Equation

mass flow rate

$$\dot{m} = \overset{\text{density}}{\rho}\,\overset{\text{area}}{A}\,\overset{\text{velocity}}{V} = \frac{AV}{\underset{\text{specific volume}}{v}}$$

$kg/s = kg/m^3 \cdot m^2 \cdot m/s$ unit: m^3/kg

ρ: fluid density
A: cross sectional area
V: velocity of fluid perpendicular to surface

2) Diameter : 0.2m velocity : 0.1 ms^{-1}
T: 25C P: 150 kPa $\dot{m} = ?$

$\Rightarrow A = \pi r^2 = 0.03 m^2$

$\Rightarrow Pv = RT$ T = 298 K

$v = \frac{RT}{P} = 0.5705\ m^3/kg$

$\dot{m} = \frac{AV}{v} = \frac{(0.03)(0.1)}{0.5705} = 0.0052\ kg/s$

4.2 Energy Equation for a Control Mass in Thermodynamics

For control mass, the energy equation is written as

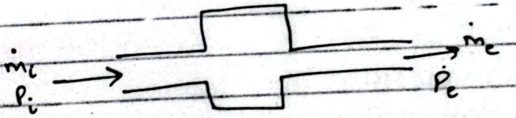
$$E_2 - E_1 = Q_{12} - W_{12}$$

Instantaneous form

$$\frac{dE_{C.M}}{dt} = \dot{Q} - \dot{W}$$

The energy eq for control volume

$$\frac{dE_{c.v}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i (e_i + P_i v_i) - \sum \dot{m}_e (e_e + P_e v_e)$$



where $e = \underbrace{U}_{\text{internal energy}} + \underbrace{\frac{V^2}{2}}_{K.E} + \underbrace{gZ}_{P.E}$

PV is work flow

Subscripts : i → incoming
e → outgoing

h_i h_e

4.2 Energy Equation for Control Volume in Thermodynamics

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i \left(U_i + gZ_i + P_i v_i + \frac{V_i^2}{2}\right) - \sum \dot{m}_e \left(U_e + \frac{V_e^2}{2} + gZ_e + P_e v_e\right)$$

The total energy for unit mass is

$$\underbrace{U + PV}_{} + \frac{V^2}{2} + gZ$$

$$\underbrace{h}_{\text{specific enthalpy}} + \underbrace{\frac{V^2}{2}}_{\text{specific K.E}} + \underbrace{gZ}_{\text{specific P.E}}$$

The energy equation for control volume

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gZ_i\right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gZ_e\right)$$

Applications:-

- Turbines
- Compressors
- Heat exchangers
- Nozzles

$W = J/s$

$\dot{W} = \frac{W}{t} = \frac{J}{s} = \text{Watt}$

$q = \frac{\dot{Q}_{cv}}{\dot{m}} = \frac{Q/t}{m/t} = \frac{Q}{m} = \frac{J}{kg}$

4.3 Steady State Processes

where the properties of system donot change with time

→ $\dot{m}_i$ and $\dot{m}_e$ are constant

→ properties of fluid entering or exiting are constant

→ Energy within control doesn't accumulate over time.

For gas turbine:

$$\underbrace{W_{c.v}}_{\text{power } (J/s)} = \dot{Q}_{cv} + \underbrace{\dot{m}\left((h_i - h_e) + \frac{V_i^2 - V_e^2}{2} + g(Z_i - Z_e)\right)}_{\text{Watt}}$$

Watt Watt

Boundary of control volume doesn't move.

- Control volume is stationary
- State of mass at each point doesn't vary with time. $\frac{dm_{cv}}{dt} = 0$, $\frac{dE_{cv}}{dt} = 0$
- Rate of heat and work remain constant.

Continuity eq. $\sum \dot{m}_i = \sum \dot{m}_e$

Energy eq: $\dot{Q}_{c.v} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gZ_i\right) = \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gZ_e\right) + \dot{W}_{cv}$

As mass flow, state, rate of heat & work constant.

Hence.

Continuity eq: $\dot{m}_i = \dot{m}_e = \dot{m}$

Energy eq:- $\dot{Q}_{c.v} + \dot{m}\left(h_i + \frac{V_i^2}{2} + gZ_i\right) = \dot{m}\left(h_e + \frac{V_e^2}{2} + gZ_e\right) + \dot{W}_{c.v}$

$$q + h_i + \frac{V_i^2}{2} + gZ_i = h_e + \frac{V_e^2}{2} + gZ_e + w$$

$$q = \frac{\dot{Q}_{c.v}}{\dot{m}} \quad \text{and} \quad w = \frac{\dot{W}_{c.v}}{\dot{m}}$$

Tayyaba Book

$\frac{Q_H - Q_L}{Q_L}$

$\frac{Q_L}{Q_H - Q_L}$

Ch#5

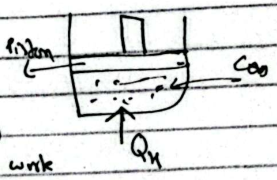
5.1 Heat Engine & Refrigarator:-

A heat engine that operates on a thermodynamic cycle, transferring heat from a high temperature to a low temp reservoirs & producing net positive work.
devices

— Steam gas or any other substance undergoes a cyclic process of heating and cooling.

Steam power plant & Engine

→ Heat Q_H is absorbed from high temp source



→ Steam flows through a turbine performing work

→ Steam is condensed, release Q_L heat to a low temp reservoir.

Thermal Efficiency:-

It is a measure of ~~how~~ how well the heat engine converts heat input into useful work

$$\eta_{thermal} = \frac{W}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

Q_L → released
Q_H → absorbed

Efficiency of real engine: $W = Q_H - Q_L$

- → Large power plant : 35-50%
- → Gasoline engine : 30-35%
- → Diesel engine : 30-40%
- → Small engine : 20%

Refrigerators & Heat Pumps:-

Refrigerator is a device that we work to transfer heat from lower temp to ~~low~~ high temp. It operates on a cycle similar to a heat pump with a focus on cooling.

Basic Cycle

- → A typical refrigerator uses a temp. cycle.
- → Refrigerator R-134a or ammonia absorbs Q_L in the evaporator. long pressure & temp
- → Refrigerator release Q_H. air as condensed.

COP

$$\beta = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L} = \frac{1}{\frac{Q_H}{Q_L} - 1}$$

Heat pumps: When the object is to heat a space rather that cool it, is called heat pump.

$$\beta = \frac{Q_L}{Q_H - Q_L} = \frac{Q_L}{W}$$

$$\beta \propto \frac{1}{\eta}$$

$$COP = \beta' = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_L} \quad \leftarrow \eta = \frac{W}{Q_H}$$

β & β'

$\frac{R_2}{R_1} \geq 2$

Second Law of Thermo.

→ Governs energy transfer and conversion in thermodynamic systems

→ Introduces concept of irreversibility and efficiency

→ The law has two classical statements

Kevin Planck statement — Clausius Statement

Kevin Planck Statement:-

"It is impossible to construct a device that operates in a cycle and produces no effect other than the raising of a weight (work done) and the exchange of heat with a single thermal reservoir."

Key implications:-

→ Heat engine cannot be 100% efficient i.e $\eta \neq 100\%$

$$\eta = \frac{W}{Q_H} \neq 100\%$$

→ Heat must flow from high to a low temp reservoir

Clausius Statement:-

"It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a cooler body to a warmer daily."

→ Refrigerator/Heat pump:- Heat cannot flow from cold reservoir to a hot reservoir without work done.

$$COP_{ref} = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L}$$

$$COP_{heat\ pump} = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_L}$$

→ Violation of Clausius statement implies violation of Kevin-Planck statement.

→ The reservoir is also true.

3- Perpetual Motion Machines

- First Kind:- Creates energy from nothing violating the 1^{st} law of thermo.
- Second Kind:- Converts heat from a single reservoir entirely into work, violating the second law $Q_H = W$
- Third Kind:- Operates indefinitely without friction or work output theoretically impossible.

Reversible Process

→ A reversible process is an idealized process that, when reversed, leaves both the system and its surroundings completely unchanged.

→ In reality, all processes are irreversible.

→ Irreversible processes deviate from equilibrium.

Factors of Irreversibility:-

1) Friction
2) Unrestrained Expansion
3) Heat transfer through a finite temp. difference
4) Mixing of different substances

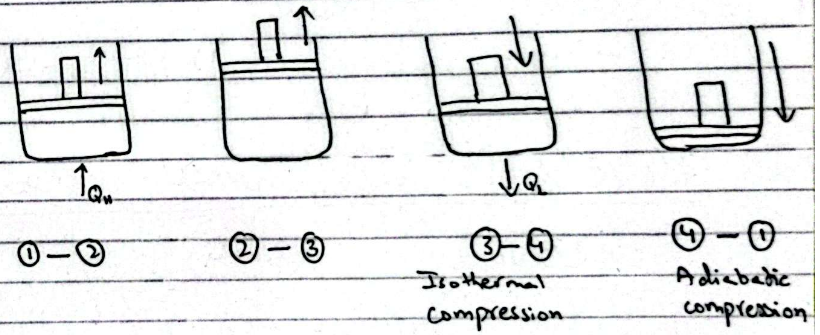
→ Internal irreversibility

→ External irreversibility

Carnot Cycle

Carnot cycle consists of 4 reversible process:-

1. Reversible Isothermal heat addition at a high temperature.
2. Reversible adiabatic expansion from the high temperature to the low temperature.
3. Reversible & isothermal heat rejector at a low temperature.
4. Reversible adiabatic compression back to the high temperature.



① Reversible Isothermal Heat Addition

- Absorbs Q_H from high temp reservoir
- Heat transfer takes place
- Working fluid expands, doing work on the surrounding
- May involve a phase change

$Q_H = T_H \Delta S$ → change in entropy

- Q and W are +ive

$U = \underset{+ive}{Q} - \underset{+ive}{W}$

(2) Reversible Adiabatic Expansion (2-3)

- Working fluid expands without exchanging heat with the surroundings
- Adiabatic process $Q=0$
- Temp of the fluid decreases from T_H to T_L
- Work is done by the system on the surroundings

$$T_H V_2^{\gamma-1} = T_L V_3^{\gamma-1}$$

volume

For ideal gas $\gamma = C_p/C_v$

- W is +ive $U = \underset{0}{Q} - \underset{+ive}{W} = -W$

(3) Reversible Isothermal Heat Rejection (3-4)

- Working fluid rejects heat Q_L to low temp. reservoir at const. temp. T_L
- Working fluid temp. is infinitesimally higher than T_L.
- The process is isothermal   Temp. const.
- Heat is removed from working fluid, causing partial condensation or compression

$$Q_L = T_L \Delta S$$

- W is -ive

(4) Reverse Adiabatic Compression (4-1)

- The working fluid is compressed adiabatically, raising back to T_H.
- Adiabatic process $Q=0$
- Reversible work done on the system by surrounding
- System temperature raise from T_L to T_H

$$T_L V_4^{\gamma-1} = T_H V_1^{\gamma-1}$$

- W is -ive $U = \underset{0}{Q} - \underset{-ive}{W} = W$

Efficiency of Carnot Cycle

$$\eta = 1 - \frac{Q_L}{Q_H} = 1 - \frac{T_L}{T_H}$$

Key features:-
- Reversible
- Depends only on the temp
- Entropy $\Delta S_H = \frac{Q_H}{T_H}$, $\Delta S_L = \frac{Q_L}{T_L}$

Applications:-

- Real world applications
- No real engine can operate on the Carnot cycle due to irreversibilities
- Carnot cycle is just a benchmark for real engines.

Two propositions regarding the efficiency of a Carnot cycle:-

1- It is impossible to construct an engine that operates b/w two given reservoirs and is more efficient than a reversible engine operating b/w the same two reservoirs.

$$\eta_{any} \leq \eta_{reversible}$$

$\eta_{any} \rightarrow$ efficiency of any engine operating b/w two reservoirs

2- All engines that operate on the carnot cycle b/w two given constant-temperature reservoirs have the same efficiency

$$\eta_{rev1} = \eta_{rev2}$$

Ch #6

Entropy

$$\oint \frac{Q}{T} \leq 0$$

The inequality applies to all possible thermodynamic cycles - whether reversible or irreversible and is used to define entropy.

Entropy is defined as

$$dS = \left.\frac{\delta Q}{T}\right|_{rev}$$

Entropy Change

$$S_2 - S_1 = \int_1^2 \left.\frac{\delta Q}{T}\right|_{rev}$$

Reversible Heat Engine Cycle:-

Net heat transfer $\quad \int \delta Q = Q_H - Q_L > 0$

Net Entropy Change $\quad \oint \frac{\delta Q}{T} = \frac{Q_H}{T_H} - \frac{Q_L}{T_L} = 0$

For all reversible cycles,

$$S_2 - S_1 = \oint \frac{\delta Q}{T} = 0$$

Irreversible Heat Engine:-

$Q_H \xrightarrow{rev} W \quad Q_L = 0$

Work $\quad W_{irr} < W_{rev}$

Heat transfer $\quad Q_L^{irr} < Q_L^{rev}$

$Q_H \quad W \quad Q_L^{irr}$

irreversible

Net entropy change

$$\oint \frac{Q}{T} = \frac{Q_H}{T_H} - \frac{Q_L^{irr}}{T_L} < 0$$

For all irreversible cycles

$$S_2 - S_1 = \oint \frac{\delta Q}{T} < 0$$

→ High entropy mean energy is scattered and thing are more disorder/randomness.

→ Low entropy energy is more organized.

When heat flows into the system, entropy increase (+ive) and vice versa

Heat into the system → +ive

Heat out of the system → -ive

Refrigerator:-

Heat transfer $\oint \delta Q = -Q_H + Q_L < 0$

Net entropy change $\oint \frac{\delta Q}{T} = \frac{-Q_H}{T} + \frac{Q_L}{T} = 0$

Irreversible Refrigeration Cycle:-

$$W_{irr} > W_{rev}$$

$$Q_H^{irr} > Q_H^{rev}$$

Net entropy change

$$\oint \frac{\delta Q}{T} = -\frac{Q_{H\,irr}}{T_H} + \frac{Q_L}{T_L} < 0$$

$\oint \frac{\delta Q}{T} = 0$ (reversible cycle)

$\oint \frac{\delta Q}{T} < 0$ (irreversible cycle)

Absolute zero → -273°C → 0 Kelvin

Entropy = 0 (Due to 3rd law of thermo)

Due to negligible movement of molecules:-

Entropy → 0

Specific Entropy (kJ/kg·K)

Saturated region:

$$S = (1-x)S_f + xS_g$$

$$S = S_f + xS_{fg}$$

Q) Water 100°C, quality 50% in a rigid box (no shape change) is heated to 110°C

How do the P, V, U, x and S changes?

(increase, decrease, same)

Ans. A fixed mass in a rigid body

V → constant $\qquad$ U ⇒ Q_{in} ↑

P → constant

Quality Increase → Change to vapour state

x → ↑

$dS \Rightarrow \frac{\delta Q \uparrow}{T} \rightarrow \uparrow$

CamScanner

Q) Determine entropy for R-410, $T = 25°C$, $V = 0.01 m^3/kg$

$x = \frac{(v - v_g)}{v_{fg}} = 0.6377$ — Table

from $s = s_f + x s_{fg} = 0.7618$ — Table

Q) Determine the missing entropy among P, T, S and x for R410a

(a) $T = -20°C$, $V = 0.1377 m^3/kg$

~~$v_f < v < v_g$~~

~~$v > v_g \rightarrow$ Superheated $x = \frac{v - v_f}{v_{fg}}$~~

~~$x \rightarrow$ undefined~~

~~(b) $P = 200 kPa$~~

~~$S = 1.4082 kJ/kgK$~~

$v_f < v < v_g$

$v > v_g \rightarrow$ superheated $\rightarrow$ undefined (quality)

$x \rightarrow$ undefine

$P = 200 kPa$, $S = 1.1783 kJ/kgK$

(b) $P = 200 kPa$, $S = 1.409 kJ/kgK$, $T = ?$, $x = ?$

$T = 60°C$, x = undefine

$S > S_g \rightarrow$ superheated

7100.591k
328.047

Q) Determine the missing property among P, V, S & $x \Rightarrow CO_2$

$T = 20°C$, $S = 1.49 kJ/kgK$

$x \rightarrow$ undefined as $S > S_g \rightarrow 1.04$
1.49 $\rightarrow$ superheated

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1} \Rightarrow \frac{y_2 - y_1}{y - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{P - 1400}{2000 - 1400} = \frac{1.49 - 1.5283}{1.4438 - 1.5283}$$

$$\boxed{P = 1671.9 kPa}$$

For V

$$\frac{1671.9 - 1400}{2000 - 1400} = \frac{V - 0.03648}{0.02453 - 0.03647}$$

$$\boxed{V = 0.03106 m^3/kg}$$